

Experiment 3

Aim: Write a program to simulate Clock Synchronization

Theory:

- Clock synchronization in distributed system refers to the process of ensuring that all clocks across various nodes or computers in the system are set to the same time or at least have their times closely aligned.
- In a distributed system, where multiple computers communicate and collaborate over a network, each computer typically has its own clock.
- However, due to factors such as hardware differences, network delays and clock drift, these local clocks can drift apart over time.
- Types:

1) Physical Clock Synchronization

- ⇒ In distributed systems each node operates with its clock, which can lead to time differences.
- However, the goal of physical clock synchronization is to overcome this challenge.
- This involves equipping each node with a clock that is adjusted to match UTC a recognized standard.
- By synchronizing their clocks in this way diverse systems, across the distributed landscape can maintain harmony.

2) Logical clock Synchronization

- ⇒ In distributed systems absolute time often takes a backseat to clock synchronization.
- Think of clocks as storytellers that prioritize the order of events than their exact timing.
 - These clocks enable the establishment of connections between events like weaving threads of cause and effect.
 - By bringing order and structure into play, task coordination within distributed systems become akin to a choreographed dance where steps are sequenced for execution.

3) Mutual exclusion synchronization

⇒ In the bustling symphony of distributed systems one major challenge is managing shared resources.

- Imagine multiple processes competing for access to the resources simultaneously.
- To address this issue mutual exclusion synchronization comes into play as an expert technique that reduces chaos and promote resource harmony.
- This approach relies on creating a system where different processes take turns accessing shared resources.

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